Heart Disease Diagnostic Tool: Technical Report

ID: 242031013

Abstract

A tuned SVM that is set to find most positive cases reaches a recall of 93.9% (31 of 33) on a small test set of 61 patients, at the cost of 11 false alarms among 28 negative patients.

The task is binary classification: from 13 clinical measurements, predict the label `target` (documented as heart disease). The data are the Kaggle copy of the UCI Heart Disease data: 1,025 rows, of which 723 are exact duplicates, which leaves 302 distinct patients. Duplicates were removed before the train/test split, so no patient is in both sets. Four models were compared: a Dummy model, Logistic Regression, a tuned SVM and a tuned Random Forest. The final model, the SVM, was chosen by cross-validated ROC-AUC on the training data (0.923), and its decision threshold (0.158) was chosen on out-of-fold training predictions with the F2 score, which weights recall twice as much as precision.

With the standard threshold of 0.5, all three real models were equal on the test set (accuracy 0.836, recall 0.848, ROC-AUC 0.898 to 0.917). In repeated cross-validation on all 302 patients the three models are also very close (ROC-AUC 0.913 to 0.915), so the complex models are not clearly better than Logistic Regression.

Important caveat about the label. In this file, the patients with `target = 1` have a higher maximum heart rate and a lower ST depression than those with `target = 0`, which is the opposite of what is expected for heart disease. The label may therefore be reversed in this copy of the data (see Data Exploration). All results below are reported for `target = 1` exactly as it appears in the file. This is a learning project and not a medical device.

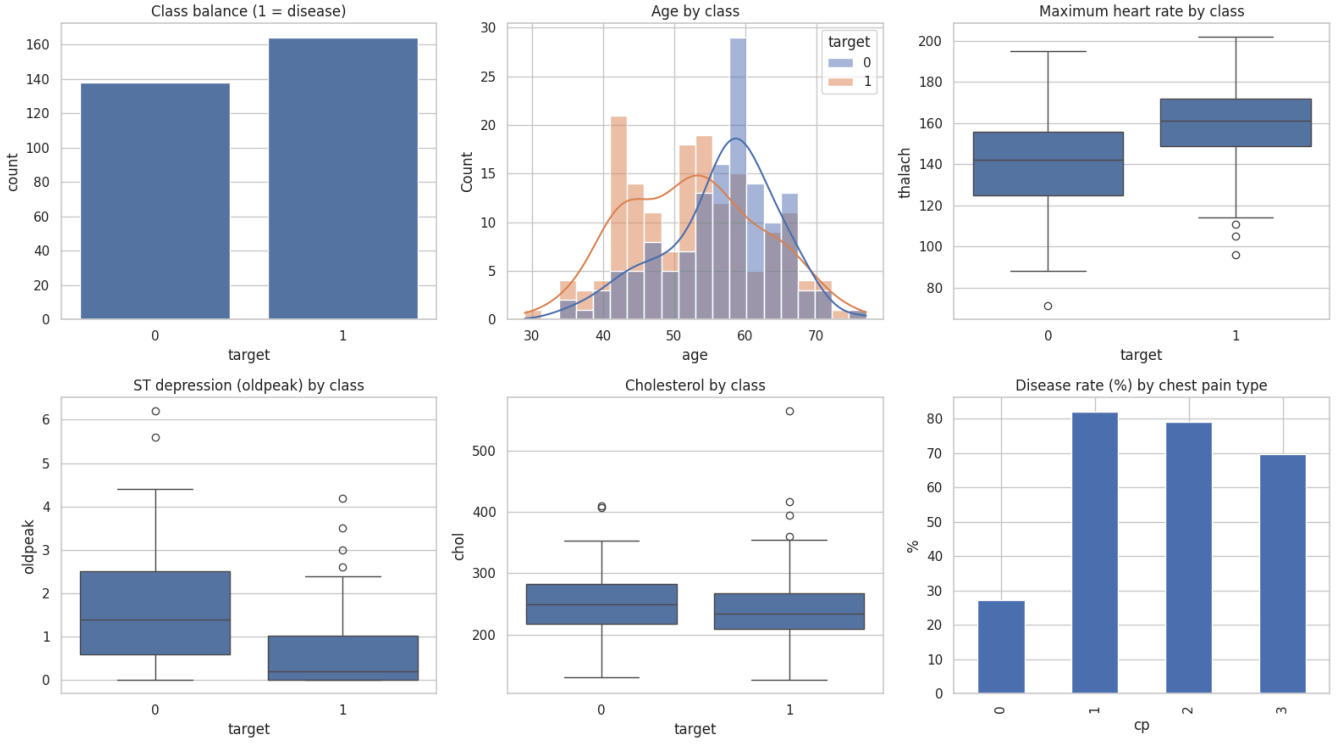
Data Exploration

The file holds only 302 distinct patients, and the label direction needs a check before the results can be interpreted medically.

Item	Value
Source	Kaggle copy of the UCI Heart Disease data (`heart.csv`)
Rows in the file	1,025 (14 columns: 13 features and `target`)

Item	Value
Exact duplicate rows	723, removed before splitting
Distinct patients	302
Missing values	none
Class counts after duplicate removal	`target = 1`: 164 (54.3%), `target = 0`: 138
Numeric features	age, trestbps, chol, thalach, oldpeak
Categorical features (stored as codes)	sex, cp, fbs, restecg, exang, slope, ca, thal

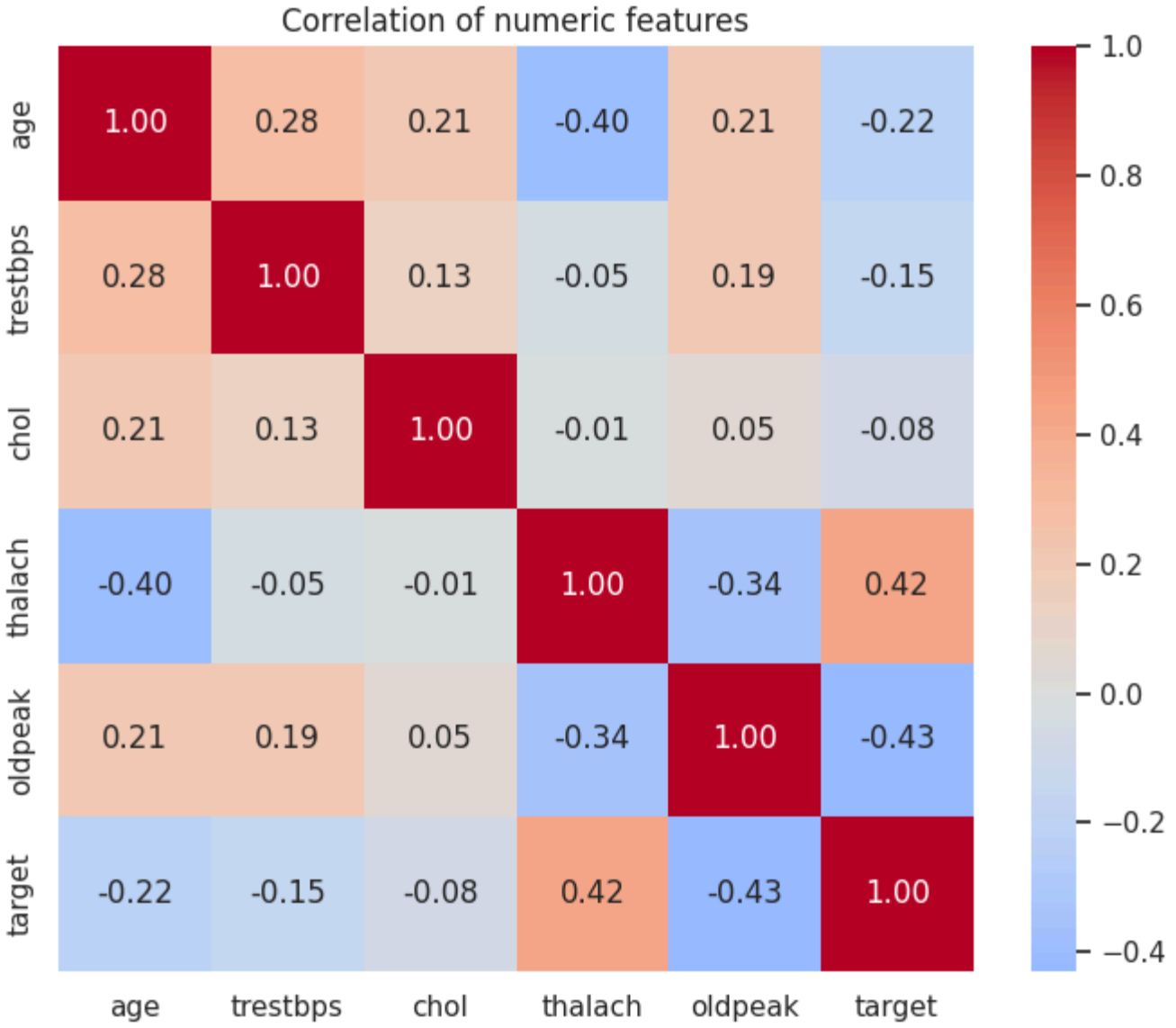
The duplicates matter: if they were kept, the same patient could appear in both the training and the test set, and the test scores would look better than they are.



What the charts show (values read from the charts are approximate)

1. **The classes are roughly balanced:** 164 patients with `target = 1` and 138 with `target = 0`.
2. **Maximum heart rate (thalach) is higher for `target = 1`:** the median is about 160 against about 142, and the correlation with the label is +0.42.
3. **ST depression (oldpeak) is lower for `target = 1`:** the median is close to 0.2 against about 1.4, and the correlation is -0.43.
4. **Age, blood pressure and cholesterol separate the classes less:** correlations with the label are -0.22 (age), -0.15 (trestbps) and -0.08 (chol).

5. **Chest pain type matters:** the share of `target = 1` is about 27% for chest pain code 0 and about 70% to 82% for codes 1 to 3. The meaning of the codes was not checked for this copy of the data.



Label check. The documentation of this Kaggle file says that `target = 1` means heart disease, but the data point the other way: a higher maximum heart rate and a lower ST depression are usually signs of a healthier heart, not of disease. In addition, the commonly cited description of the original UCI Cleveland data lists 164 patients without disease and about 139 with disease, and this file has 164 patients with `target = 1`. I have not verified that UCI count against the UCI page, so this is a strong hint and not proof. If the label is reversed in this copy, `target = 1` means "no heart disease" and every recall and precision value in this report describes the healthy class. The notebook has a setting (`INVERT_TARGET`) that flips the label, but the results below were produced without flipping it.

Methodology

Duplicates are removed first, all preprocessing is learned from the training patients only, and the threshold is chosen without looking at the test set.

Preprocessing (inside pipelines)

- Numeric features: median imputation, then standard scaling (scaling for Logistic Regression and SVM only; the Random Forest uses unscaled values).
- Categorical features (stored as codes): most-frequent imputation, then one-hot encoding.
- Because these steps sit inside each pipeline, they learn only from the training part of every split.

Split

Set	Patients	Share with `target = 1`
Train (80%, stratified, random seed 42)	241	54.4%
Test (20%)	61 (33 with `target = 1`)	54.1%

Models

Model	Role
Dummy	Always predicts the most frequent class (the minimum to beat)
Logistic Regression	Simple baseline
SVM (tuned)	Complex model: RBF or linear kernel, probability outputs
Random Forest (tuned)	Complex model: many decision trees voting together

Hyperparameter tuning

- RandomizedSearchCV with 25 random trials per model, 5-fold stratified cross-validation on the training patients only, scored by ROC-AUC.

- SVM search space: C from 0.01 to 100 (20 log-spaced values), gamma from 0.0001 to 1 (20 log-spaced values), kernel rbf or linear. Best: rbf kernel, C of about 14.4, gamma of about 0.0011, cross-validated ROC-AUC 0.923.
- Random Forest search space: 100 / 200 / 300 / 500 trees, `max_depth` 3 / 4 / 5 / 6 / 8 / none, `min_samples_leaf` 1 / 2 / 4 / 8, `max_features` sqrt / log2 / 0.5. Best: 100 trees, depth 6, leaf size 8, log2 features, cross-validated ROC-AUC 0.917.
- **Final model:** the SVM, because it has the higher cross-validated ROC-AUC on the training data. The test set was not used for this choice.

Decision threshold

Missing a patient who has the condition is worse than a false alarm, so the threshold was not left at 0.5. It was set to the value that maximises the F2 score (recall counts twice as much as precision) on out-of-fold predictions of the training data. The chosen threshold is 0.158 (out-of-fold F2 = 0.923). The test set was not used.

Metrics

- Recall: of the patients with `target = 1`, the share that the model finds (the main metric).
- Precision: of the patients the model flags, the share that really have `target = 1`.
- F1 and ROC-AUC: overall balance of the two and the quality of the ranking.
- Because the test set is small, the models were also compared with 5-fold cross-validation repeated 5 times on all 302 patients. The tuned settings came from the training split, so these cross-validation numbers are slightly optimistic for the tuned models and are used to compare models with each other.

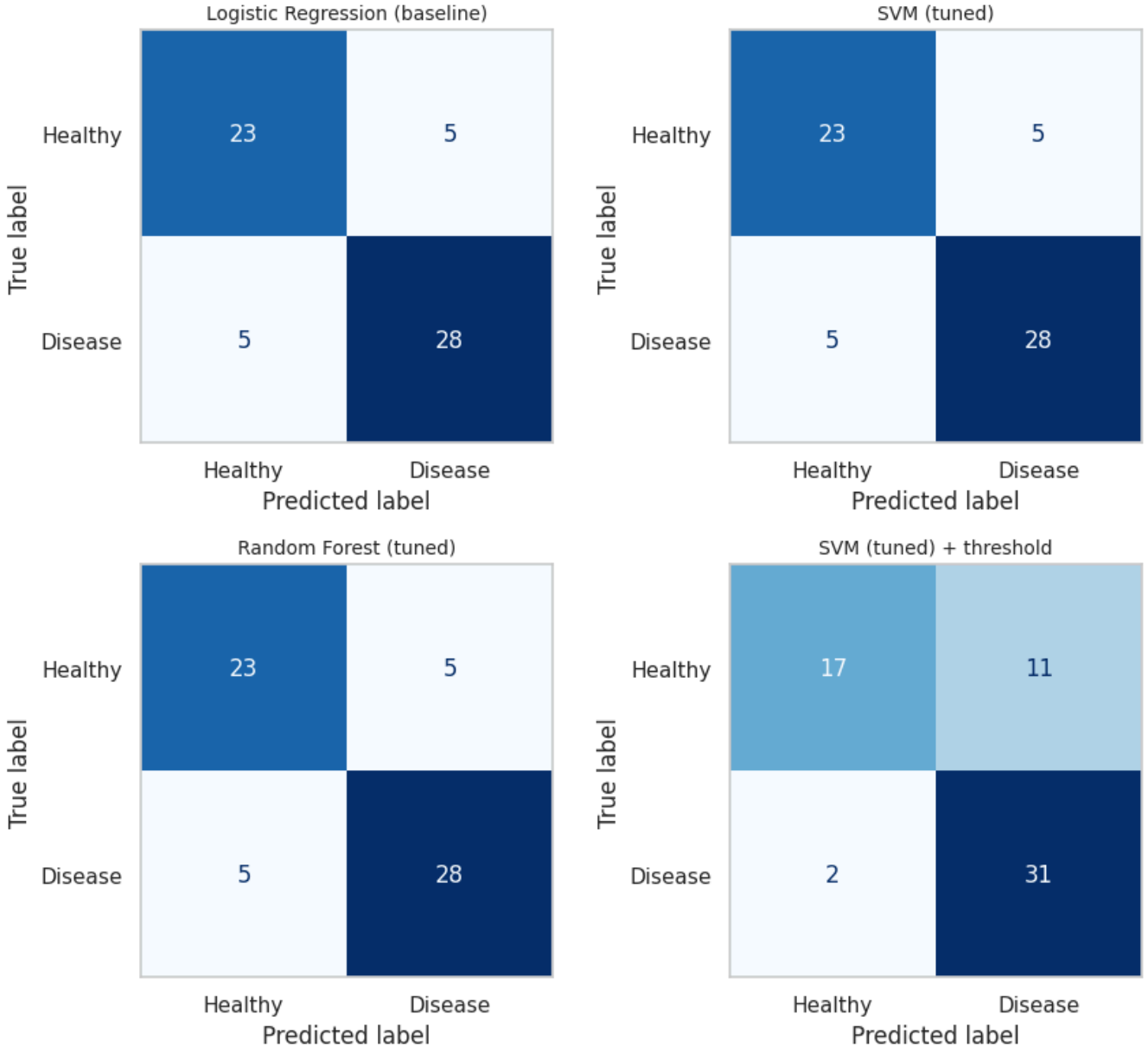
Results

Lowering the threshold raised recall from 0.848 to 0.939 on the 61 test patients, and the three real models are otherwise equal.

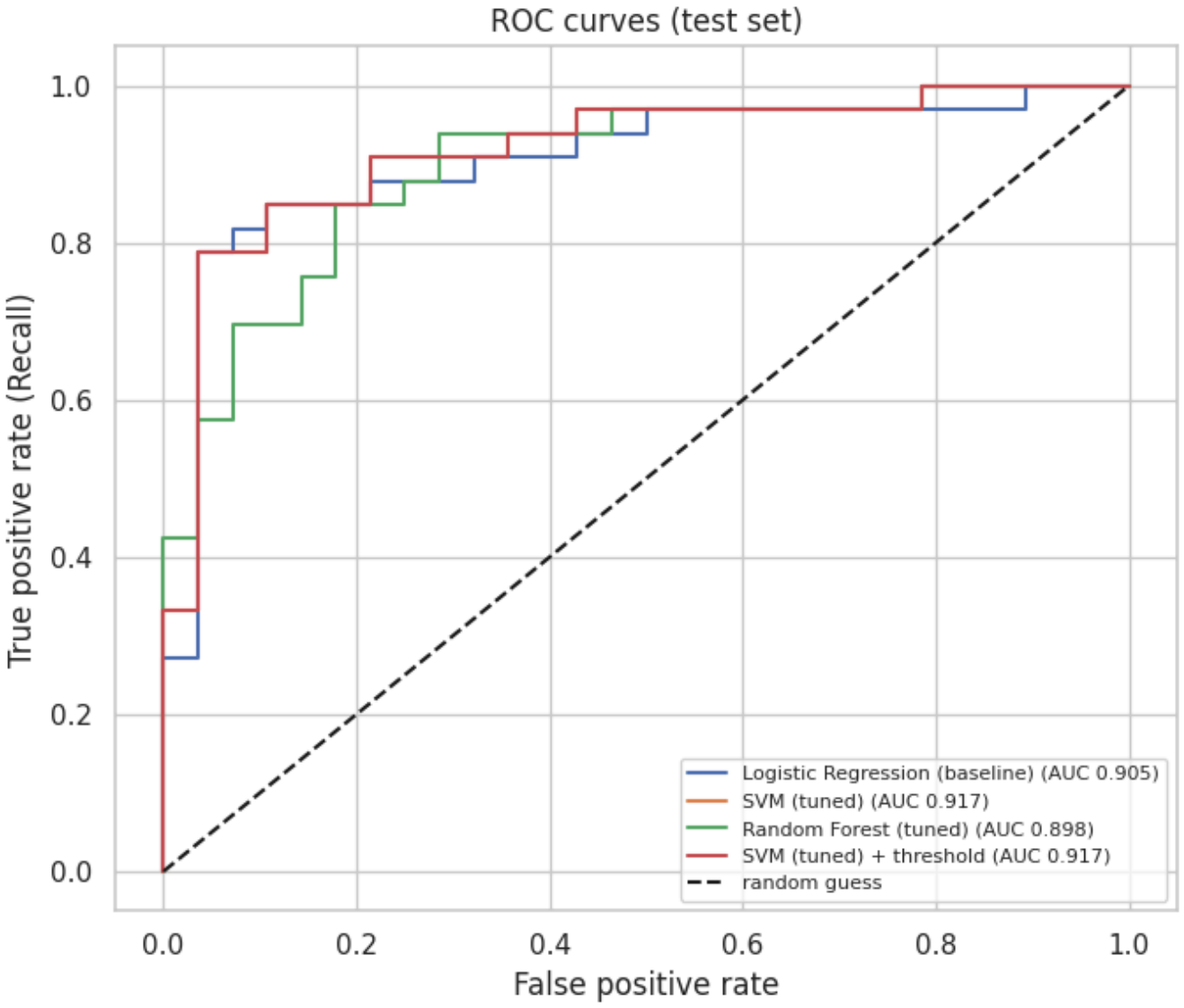
Test set (61 patients, 33 with `target = 1`)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Dummy (most frequent)	0.541	0.541	1.000	0.702	0.500
Logistic Regression (baseline)	0.836	0.848	0.848	0.848	0.905
SVM (tuned)	0.836	0.848	0.848	0.848	0.917
Random Forest (tuned)	0.836	0.848	0.848	0.848	0.898
SVM (tuned) + threshold 0.158 (final)	0.787	0.738	0.939	0.827	0.917

The Dummy model gets a recall of 1.000 only because it flags every patient as `target = 1`; its precision of 0.541 equals the share of that class.



- **At the standard threshold of 0.5** the three real models make the same number of mistakes: 28 of 33 positives found (5 missed) and 5 of 28 negatives flagged by mistake. They differ only in ROC-AUC (0.898 to 0.917).
- **With the threshold of 0.158** the final SVM finds 31 of 33 positives (recall 0.939, 2 missed) but flags 11 of 28 negatives by mistake (precision 0.738). Accuracy falls from 0.836 to 0.787 and F1 from 0.848 to 0.827. ROC-AUC stays 0.917, because the threshold does not change the ranking.
- **Small test set.** One patient is worth about 3 percentage points of recall (1 of 33), so a difference of one or two patients between models is not reliable.

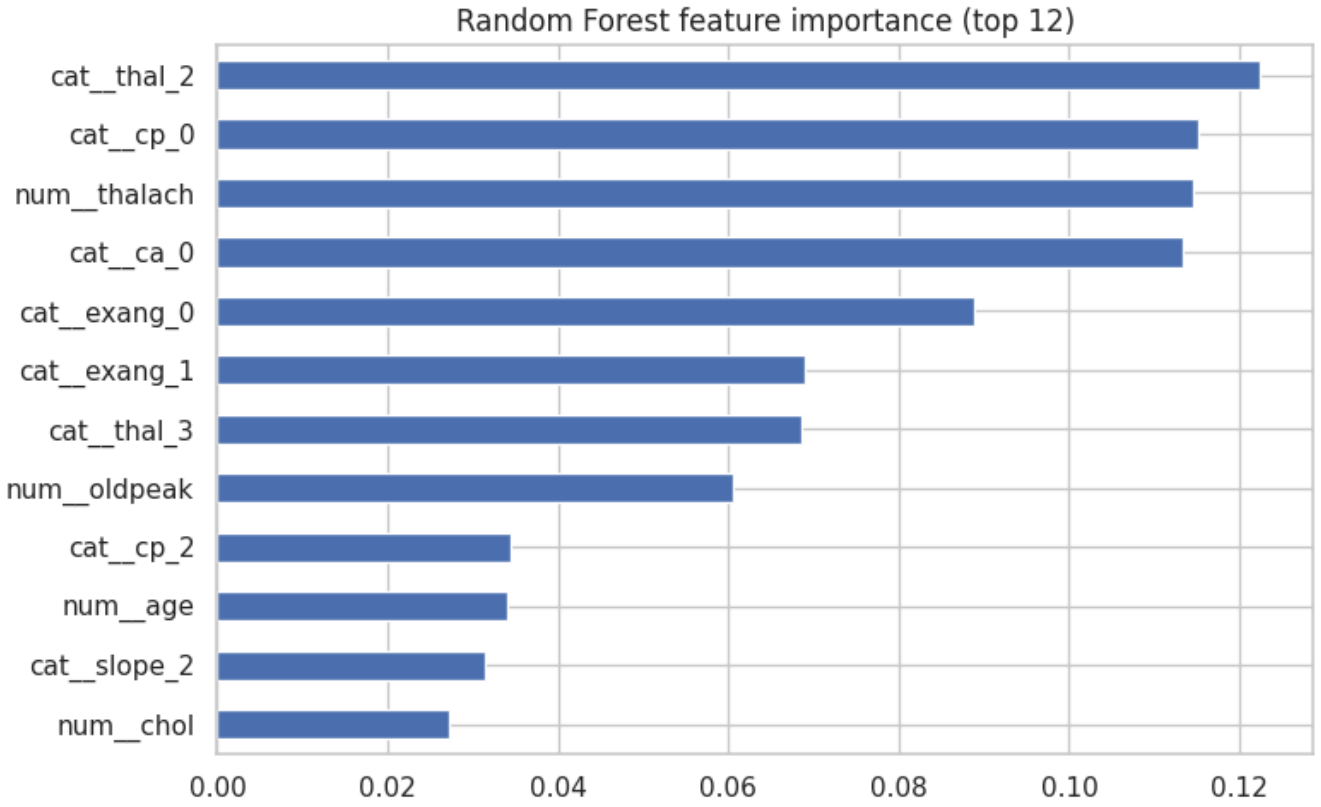


The ROC curves of the models lie close to each other. The curve of the SVM and of the SVM with threshold are the same curve (only the cut-off point moves), so one is hidden behind the other in the chart.

Repeated stratified cross-validation (5 folds, 5 repeats, all 302 patients, mean ± standard deviation)

Model	Recall	Precision	F1	ROC-AUC
Dummy	1.000 ± 0.000	0.543 ± 0.006	0.704 ± 0.005	0.500 ± 0.000
Logistic Regression	0.883 ± 0.063	0.845 ± 0.056	0.861 ± 0.035	0.914 ± 0.030
SVM (tuned)	0.879 ± 0.068	0.833 ± 0.045	0.852 ± 0.028	0.915 ± 0.023
Random Forest (tuned)	0.856 ± 0.061	0.829 ± 0.047	0.840 ± 0.028	0.913 ± 0.026

The three real models are within one standard deviation of each other on every metric, and Logistic Regression is as good as the two complex models.



Feature importance. The chart uses the Random Forest, because the SVM has no built-in importances. The top features are thal code 2, chest pain code 0, maximum heart rate (thalach), ca code 0 and exang code 0, each with an importance of roughly 0.09 to 0.12 (read from the chart). The category names are one-hot codes, and their medical meaning was not checked for this copy of the data. Importances of correlated features share their values.

Conclusion

On this small dataset a simple Logistic Regression is about as good as a tuned SVM or Random Forest, and the choice of threshold changes the result more than the choice of model.

Model comparison. All three real models beat the Dummy model clearly (ROC-AUC 0.898 to 0.917 on the test set, 0.913 to 0.915 in repeated cross-validation, against 0.500). They are not clearly different from each other. The SVM was chosen as the final model only because of its slightly higher cross-validated ROC-AUC on the training data (0.923 against 0.917 for the Random Forest).

Trade-off. The threshold of 0.158 finds 31 of 33 positive patients (2 missed) instead of 28 of 33 (5 missed), but 11 of 28 negative patients are flagged by mistake instead of 5. Whether this is acceptable depends on the cost of a missed case against the cost of an extra check, which this project does not know.

Limitations

- **The label may be reversed.** The patients with `target = 1` have a higher maximum heart rate and lower ST depression, which is the opposite of what is expected for heart disease, and the class counts match the UCI counts in reverse order (not verified against the UCI page). If so, `target = 1` means no heart disease and the recall in this report describes the healthy class. The notebook can flip the label with `INVERT_TARGET = True`; the results here were produced without flipping it.
- Only 302 distinct patients and a test set of 61 (33 with `target = 1`): one patient moves recall by about 3 percentage points.
- The tuned settings and the threshold came from the training split, so the repeated cross-validation numbers are slightly optimistic for the tuned models.
- The data come from one source population and were not validated on an outside hospital.
- The meaning of the coded categorical columns was not checked for this copy of the file.
- This is a learning project. It is not a medical device and must not be used to diagnose patients.

Future work

- Check the label direction against the original UCI data, and re-run the notebook with the correct label.
- Collect more patients and test the model on an external dataset.
- Calibrate the predicted probabilities and choose the threshold from real costs of missed cases and extra checks.
- Try feature selection and gradient boosting, and serve the model through a FastAPI endpoint.

Reproducibility. Code: `project4_heart_disease_diagnostic.ipynb` (random seed 42). Python libraries: pandas, scikit-learn, matplotlib and seaborn.

Source of the data: Kaggle copy of the UCI Heart Disease data (`heart.csv`).